SensiSpace

College Demo Script

Satellite Land Cover Classification

Baby Dragon Hatchling vs ResNet-18

Total Demo Time: ~12-15 minutes

Demo Flow Overview:

Time	Step	What You Do
Before	Setup	Start server, open browser
2 min	Introduction	Problem + your solution
1 min	Dataset	Show EuroSAT samples, 10 classes
3 min	Live Demo	Classify 2-3 images, show tabs
4 min	Architecture	ResNet + BDH 4 modules (KEY!)
1 min	Visualization	Explain spectral segmentation
1 min	Conclusion	Results + future work

STEP 0: Setup (Before Your Turn)

Do this 5 minutes before your demo starts.

[ACTION] Open Terminal and run:

```
cd "/Users/yashlohia/Major project"
source venv/bin/activate
PYTHONPATH=$(pwd)/src uvicorn app:app --app-dir src --host 0.0.0.0 --port 8000
```

[ACTION] Open Chrome -> http://localhost:8000

[ACTION] Keep the PDF report open as backup: SensiSpace_Project_Report.pdf

Load 2-3 diverse samples beforehand (forest, city, river) so you know what to expect.

STEP 1: Introduction (2 minutes)

What to say:

"Good morning everyone. My project is called SensiSpace - a Satellite Land Cover Classification System.

The problem: Organizations like ISRO collect millions of satellite images every day. Manually analyzing each image to identify what type of land it shows - is it a forest? a river? residential area? - is impossible at this scale. We need AI to do this automatically.

My solution: I have built a deep learning system that takes any satellite image as input and instantly classifies it into one of 10 land cover types - Forest, River, Highway, Residential, Agricultural crops, and so on.

What makes this unique: I designed a completely new neural network architecture called Baby Dragon Hatchling, inspired by how the human brain processes visual information. I compare it against ResNet-18, a standard industry CNN.

The system is trained on EuroSAT - 27,000 real satellite images from the Sentinel-2 satellite."

STEP 2: Show the Dataset (1 minute)

[ACTION] Click "Load Random EuroSAT Sample" a few times to show different images.

"These are actual 64x64 pixel satellite images from space. Each one is labeled with one of 10 classes:
- Forest: dense tree cover
- River: water bodies
- Residential: houses and neighborhoods
- Highway: major roads
- AnnualCrop: agricultural fields
- And 5 more types.

We train on 21,600 images and test on 5,400."

STEP 3: Live Demo (3 minutes)

This is the WOW moment. Pick interesting images.

[ACTION] Click "Load Random EuroSAT Sample" - pick one with visible features

[ACTION] Click "Run Classification" - wait ~2 seconds

While loading, say:

"Now I am running both models on this image simultaneously..."

Once results appear:

"Here you can see the results. ResNet-18 classified this as [read prediction] with [read confidence]% confidence in [read time] milliseconds.

Below that is the spectral land cover map - pixel-level highlighting showing exactly WHICH parts of the image correspond to which land type. Notice how the boundaries follow actual features in the image."

[ACTION] Click the "Baby Dragon Hatchling" tab

"My custom model - Baby Dragon Hatchling - classified this as [read prediction] with [read confidence]% confidence.

Look at the probability distribution chart - every class is color-coded."

[ACTION] Click the "Comparison" tab

"This comparison view shows both models side by side."

Repeat with 2-3 different samples for variety.

STEP 4: Explain Architecture (4 minutes)

THIS IS WHERE THE MARKS ARE. Speak clearly and confidently.

Part A: ResNet-18

"ResNet-18 is a well-known CNN - 18 layers of 3x3 convolutions with skip connections. The skip connections solve the vanishing gradient problem by adding the input directly to the output of each block. We use it pretrained on ImageNet - 1.3 million natural images - and fine-tune on satellite data. It has about 11.2 million parameters."

Part B: Baby Dragon Hatchling (THE KEY)

"Baby Dragon Hatchling is my novel architecture, inspired by neuroscience. It has only 1.8 million parameters - 6 times fewer than ResNet - but much higher accuracy. It has 4 biologically-inspired modules:"

Module 1: Multi-Scale Feature Pyramid

"Instead of one 3x3 convolution per layer, BDH uses three parallel convolutions: 3x3, 5x5, and 7x7 simultaneously. The small kernel captures fine details like road edges. The medium captures buildings. The large captures broad patterns like forests. All three are concatenated. This is critical because in satellite images, a house is tiny but a forest is huge."

Module 2: Lateral Inhibition

"From neuroscience: in your retina, when a neuron fires, it suppresses neighbors. This sharpens edge perception. BDH does the same - a depthwise convolution estimates surrounding activity, then we subtract a learnable fraction (alpha) from the center. This sharpens boundaries between land types - like where a river meets a forest."

Module 3: Contextual Attention

"A dual attention mechanism. Channel Attention asks WHAT features matter using squeeze-and-excitation. Spatial Attention asks WHERE to look using pooling and 7x7 convolution. Together they give selective focus - like how you immediately notice the river in a complex satellite photo."

Module 4: Recurrent Refinement

"The brain does not process images in one shot - it has feedback loops. BDH runs 2 iterations of gated refinement. Each iteration proposes an improvement, and a sigmoid gate decides how much to accept. Features get progressively cleaner - like how you notice more details the longer you look at something."

The Punchline (say this with emphasis):

"The result: 95.87% accuracy with BDH vs 81.98% with ResNet-18. A 14 percentage point improvement with 6x fewer parameters. This proves that domain-specific, biologically-inspired design beats brute-force parameter scaling."

STEP 5: Visualization (1 minute)

"For the visualization, instead of using Grad-CAM which produces blurry blobs, I use spectral color-space analysis - the same technique used in real satellite remote sensing.

It converts the image to HSV and LAB color spaces, computes a Green Leaf Index to identify vegetation, identifies water by blue hue, and urban areas by warm gray tones. This gives pixel-accurate boundaries that follow actual land features. The model probabilities are used to label each region with the correct class."

STEP 6: Conclusion (1 minute)

"The tech stack: PyTorch for deep learning, FastAPI for the backend API, and vanilla HTML/CSS/JavaScript for the frontend.

In conclusion: This project demonstrates that biologically-inspired neural architectures can significantly outperform standard CNNs for satellite remote sensing. Baby Dragon Hatchling achieves near state-of-the-art accuracy with a fraction of the parameters, while providing interpretable, pixel-level land cover mapping.

Thank you. I am happy to take questions."

ANTICIPATED QUESTIONS & ANSWERS

Q: What is Baby Dragon Hatchling? Why that name?

A: It is a novel neural network architecture I developed for this project. The name reflects that it is a small but powerful model - like a baby dragon - designed to grow into larger applications. It is referenced in the context of ISRO RESPOND initiative for intelligent Earth observation.

Q: Why is BDH better than ResNet?

A: Three reasons: (1) Multi-scale processing captures objects at all sizes, (2) Lateral inhibition sharpens boundaries between land types, (3) Contextual attention lets it focus on what matters. ResNet only has uniform 3x3 convolutions with no attention or inhibition.

Q: What is lateral inhibition?

A: A biological mechanism in the retina. When a neuron fires, it suppresses neighbors, creating sharper edge perception. In BDH, a depthwise convolution estimates surrounding activity, then we subtract a learnable fraction from the center response. This amplifies differences between adjacent features.

Q: Why not use a bigger model like ResNet-50 or ViT?

A: The point is to show that intelligent architecture design beats brute force. BDH has 1.8M parameters vs ResNet-18 11.2M, yet outperforms it by 14%. A bigger model uses more compute without addressing the fundamental limitations of single-scale, attention-less processing.

Q: What dataset did you use?

A: EuroSAT - 27,000 satellite images from the Sentinel-2 satellite, 64x64 pixels each, 10 land cover classes. It is a standard benchmark in remote sensing research.

Q: Why spectral segmentation instead of Grad-CAM?

A: Grad-CAM works on classification models that predict one label for the whole image. It produces blurry activation blobs, not precise boundaries. Spectral segmentation analyzes actual pixel colors - green for vegetation, blue for water - giving pixel-accurate boundaries. Same principle as professional remote sensing NDVI indices.

Q: Can this work on real-time satellite feeds?

A: The inference time is under 30ms per image, so yes. The current system processes one image at a time via the web interface, but the architecture could be deployed for batch processing of satellite data streams.

Q: What is the recurrent refinement doing exactly?

A: After feature extraction, 2 iterations where: (1) propose refined features via convolution, (2) compute a gate value 0-1 for each position, (3) blend original and refined based on the gate. Where gate is high, keep original. Where low, use refined. Features become progressively cleaner.

Q: How did you train it?

A: Adam optimizer, learning rate 0.001 with StepLR decay, CrossEntropy loss, batch size 64, 15 epochs with data augmentation (random flips, rotations, color jitter). Both models trained on the same 80/20 split for fair comparison.

Q: What is the scope / future work?

A: (1) Train a semantic segmentation model (U-Net/DeepLab) for true pixel-level classification, (2) Support higher-resolution imagery from ISRO satellites, (3) Deploy as a cloud service for real-time Earth monitoring, (4) Apply to deforestation tracking, flood mapping, or urban expansion.

PRO TIPS FOR THE DEMO

- Load 2-3 diverse samples BEFORE your demo starts - forest, city, river
- Speak SLOWLY when explaining the 4 BDH modules - that is where the marks are
- Point at the screen when showing spectral overlay - "notice how the green follows the tree line"
- If a classification is wrong, say: "this shows the confidence distribution - even when wrong, the correct class usually appears in the top 3"
- Keep the PDF report open as backup: SensiSpace_Project_Report.pdf
- Make eye contact with the evaluator, not the screen
- End strong with: "95.87% accuracy, 6x fewer parameters, biologically inspired"

RECOVERY COMMANDS

If anything goes wrong, use these:

Start the server:

```
cd "/Users/yashlohia/Major project"
source venv/bin/activate
PYTHONPATH=$(pwd)/src uvicorn app:app --app-dir src --host 0.0.0.0 --port 8000
```

Revert to saved state:

```
git checkout returnyash
```